

Utilizing a Conversational Agent to Promote Self-efficacy in Children

A Pilot Study on Low Cognitive Ability Children with Attention Deficit Hyperactivity Disorder

Doeun Park
HCI Lab, Yonsei University Seoul,
South Korea
doeun14@naver.com

Myounglee Choo
HCI Lab, Yonsei University Seoul,
South Korea
myounglee.choo@gmail.com

Bohyun Jin, M.D.
Department of Psychiatry,
Kyungpook National University
Children's Hospital Daegu, South
Korea
j_constellation@hanmail.net

Un Sun Chung, M.D., Ph.D.
Department of Psychiatry, School of
Medicine Kyungpook National
University Daegu, South Korea,
Kyungpook National University
Children's Hospital Daegu, South
Korea
unsunchung@gmail.com

Jinwoo Kim, Ph.D.
HCI Lab, Yonsei University Seoul,
South Korea
jinwoo@yonsei.ac.kr

Junghan Lee, M.D., Ph.D.
Department of Psychiatry, College of
Medicine Yonsei University Seoul,
South Korea, Severance Hospital
Seoul, South Korea
humanjhl@yuhs.ac

Yee-Jin Shin, M.D., Ph.D.
Department of Psychiatry, College of
Medicine Yonsei University Seoul,
South Korea, Severance Hospital
Seoul, South Korea
yjshin@yuhs.ac

ABSTRACT

Self-efficacy is important for children trying to learn new things. However, children with attention deficit hyperactivity disorder (ADHD), mild intellectual disability (MID), or borderline intellectual function (BIF) have less experience with success and lower self-efficacy compared to other children. This study is based on the hypothesis that the self-efficacy of children is enhanced by independently practicing and completing daily tasks. This study aims to promote self-efficacy in low cognitive ability children with ADHD. We developed a conversational agent that helps children initiate and complete their daily tasks. A pilot study was conducted in which ten children with 1) ADHD and 2) MID or BIF used the conversational agent for eight weeks. The participants exhibited an average compliance rate of 76.1%. Statistical analysis of the survey results revealed improvements in children's self-efficacy and ADHD symptoms. This pilot study discusses new possibilities for conversational agents for low cognitive ability children with ADHD.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Conversational Agent, Assistive Technology, Attention Deficit Hyperactivity Disorder (ADHD), Mild Intellectual Disability (MID), Borderline Intellectual Functioning (BIF)

ACM Reference Format:

Doeun Park, Myounglee Choo, Bohyun Jin, M.D., Un Sun Chung, M.D., Ph.D., Jinwoo Kim, Ph.D., Junghan Lee, M.D., Ph.D., and Yee-Jin Shin, M.D., Ph.D. 2023. Utilizing a Conversational Agent to Promote Self-efficacy in Children: A Pilot Study on Low Cognitive Ability Children with Attention Deficit Hyperactivity Disorder. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585887>

1 INTRODUCTION

Self-efficacy is the belief in one's capabilities to organize and execute the courses of action required to manage prospective situations [3]. Bandura mentioned that this belief determines how people think, act, and feel [2]. Moreover, self-efficacy is important for children when facing challenges and learning new things. Therefore, considerable research has been conducted in education and development [8, 9].

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585887>

Intellectual disability (ID) is a neurodevelopmental disorder characterized by both intellectual and adaptive function deficits in conceptual, social, and practical domains [5]. Children with mild intellectual disability (MID), who have an intelligence quotient (IQ) between 50 and 70, can undergo social and professional rehabilitation through training [5]. Many children with MID attend general elementary schools for children with typical development. However, children with MID have more problems with emotion and sociability and they are less intelligent than average, which result in more failures [31]. Additionally, the self-efficacy of MID tends to be low [34]. Borderline intellectual function (BIF) is characterized by an IQ between 71 and 84, accounting for up to 18% of the general population [28]. BIF is not a disability or a disease, but it is difficult to acquire basic academic skills because of low cognitive ability [6, 35]. Therefore, children with BIF experience educational, personal, and social problems from an early age [6]. In particular, while attending school, they are rejected by their peers and have difficulty with social adaptation [18]. Therefore, children with BIF tend to have low self-confidence and self-efficacy [7].

Children with MID find it difficult to plan and work alone, even when considering daily tasks such as organizing things and managing time. Such challenges are related to executive function, which involves adapting to and managing thoughts and behaviors to meet individual goals [38]. Children with MID have a significant deficiency in executive function compared with typically developing children of the same age [22]. Attention deficit hyperactivity disorder (ADHD) is a representative disease associated with defects in executive function [37]. Children with ADHD are likely to frequently fail tasks, resulting in a negative self-image and low self-efficacy [33]. Children with ID are at an elevated risk of being diagnosed with ADHD [10].

This study aims to enhance the self-efficacy of low cognitive ability children with ADHD using a developed conversational agent. The conversational agent can aid children's completion of daily tasks, and mastery experience promotes self-efficacy [23]. Through repeated mastery experiences, the conversational agent promotes self-confidence in children and subsequently enhances their self-efficacy.

2 RELATED WORK

2.1 Assistive Technology for Improving Self-efficacy in ADHD and ID

Self-efficacy is the belief in the possibility of success that an individual can successfully perform a given task or action [3]. Therefore, people with similar capabilities may have different attempts, performances, and task performance results depending on their self-efficacy levels. Although there are other factors, active mastery experiences are known to significantly affect the formation of self-efficacy [4, 30]. However, children with neurodevelopmental disorders, such as ADHD and ID, with relatively few successful experiences have low self-efficacy. Self-efficacy plays a significant role when individuals take action. This is because self-efficacy influences individuals' behavior initiation, performance, and continuation. Therefore, research on behavior change often seeks to enhance self-efficacy [1].

Some studies have attempted to enhance self-efficacy using assistive technology to induce positive behavior in children with ADHD and ID. A study that helps children with ADHD with daily life planning and management with smartwatches also explored improvements in self-efficacy [26]. Similarly, a study was conducted to improve the self-efficacy of children with ADHD by managing daily tasks through a voice-bot [12, 13]. A study promoted the physical activity of children with ID using activity trackers to improve self-efficacy [21]. However, these studies were limited to protocols or plans, and their effectiveness was not adequately understood.

This study is also a pilot study targeting a small number of children. However, we attempted to present the research results with various quantitative and qualitative data collected from children who used assistive technology and their parents.

2.2 Self-instruction Training

Self-instruction is a cognitive behavioral therapy (CBT) method. CBT interventions aim to change children's thinking, leading them to change their behaviors independently [14]. Self-instruction allows a child to self-execute by reciting the order in which they will perform their actions [16]. Thus, self-instruction provides a strategy that enables children to actively use self-regulation in new situations [19]. These experiences cause children to act voluntarily and perceive their actions as subjective, which is a benefit of self-instruction.

This study used a method similar to previous studies in which intervention was delivered to children with ADHD by applying self-instruction training to the dialog of a voice-bot [12, 13]. Self-instruction is a non-pharmacotherapeutic method applied to people with ADHD [36]. Furthermore, many studies have also applied self-instruction to people with MID [25, 29]. Therefore, this method may be suitable for children with both MID and ADHD.

3 METHOD

3.1 Prototype

We developed an application with a chatbot builder to allow children with ADHD and a conversational agent to converse (Figure 1) [15]. The conversational agent assists children with ADHD in starting, performing, and completing daily tasks that they need to do themselves. The previous study confirmed whether the prototype was an appropriate tool for children through a usability test [15].

Our initial study reported that children with ADHD continuously interacted with a voice-bot to complete daily tasks to be performed [13]. This study utilized the scenario developed in the study [12]. Furthermore, verbal scaffolding elements and the dialog structure of Goal-Plan-Do-Check, which is based on the self-instruction training structure, were used.

Considering the ADHD characteristics of the participating children, the agent verbally asked the child to complete tasks by following a detailed process divided into several procedures. For example, the "hand wash" task is divided into three procedures: 'rolling up one's sleeves,' 'washing one's hands for 30 s,' and 'wiping one's hands with a towel.' A customized procedure was provided in three grades, according to the child's Korean version of the ADHD rating scale (K-ARS) score. The higher the K-ARS score, the more detailed the daily tasks. For children with a K-ARS score of 24 points or

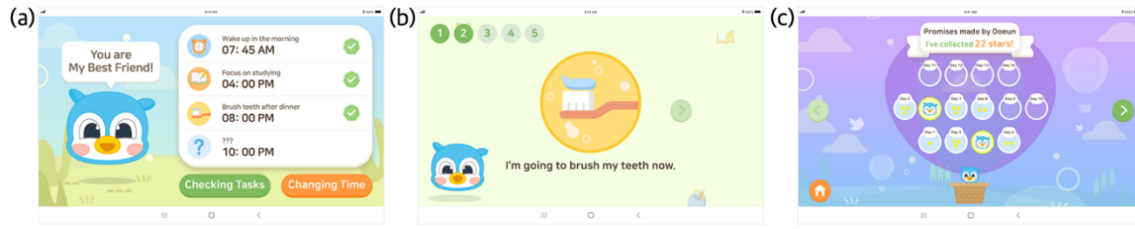


Figure 1: (a) Main page where a child can check whether a task has been executed. (b) The output screen shown when a task has been performed. Using speech-to-text, the child's voice appears on the screen. (c) A screen that allows the child to check the task execution results. One star is awarded for each completed task. If all the tasks for the day are complete, the stars become a sticker.

higher, "hand wash" was performed in three procedures as in the example above, and 'washing one's hands for 30 s' was performed with one procedure for children with a score of fewer than 17 points.

At the time specified by the child, the agent provided a reminder asking, "What do you need to do now?" The reminder helps the child to start the task that they are supposed to do. Alternatively, the child could press the tablet to perform daily tasks.

Tasks comprise 16 activities that lower-grade children in elementary school need to do independently, including getting up on their own, eating by themselves without wandering, organizing their coats, and studying. In this study, the children (after consulting their parents) chose to perform three tasks for a week. The children were allowed to adjust the time needed to perform the tasks or choose not to perform the tasks every day. Each day, the agent asked the child what happened during the day (e.g., "what was the most pleasant thing you did today?" or "what was new today?") Thus, the child interacted with the agent up to four times daily.

3.2 Study Design

With the cooperation of the education office, an announcement was sent to six elementary schools to recruit the participants. This study was conducted with the consent of the children and their parents. The participants were selected according to three criteria. Children 1) aged six–seven years, 2) diagnosed with ADHD by a pediatric psychiatrist per DSM-5 and K-SADS-PL, and 3) diagnosed with MID or BIF using K-WISC-V.

This study adopted a pretest-posttest design. For eight weeks, the children used the prototype on a tablet PC that the researcher distributed to each household on the first day of the study. On the first and last days of the study, a comprehensive attention test (CAT) was conducted on the children at a university hospital. The researchers visited homes on the first day of the study and on weeks 4 and 8 to conduct the K-ARS and self-efficacy scale surveys and interviews with the children and their parents.

3.3 Outcome Measures

U and Shin developed a self-efficacy scale for children with IDs and demonstrated its reliability and validity [32]. This scale was based on the general self-efficacy scale developed by Sherer et al. [27]. This tool comprises three factors: (1) self-regulatory efficacy (the degree to which a person can self-regulate when completing tasks), (2) self-confidence (the extent to which one believes they can

complete a task), and (3) task difficulty preference (the degree to which a person prefers when setting goals). It consists of 15 items, and the children directly answered the survey.

The K-ARS measures ADHD symptoms. This scale is the Korean version of the ADHD rating scale developed by DuPaul et al. [17]. The reliability and validity of the K-ARS has been confirmed [39]. This tool evaluates two factors: 1) inattention and 2) hyperactivity-impulsivity.

CAT is a type of continuous performance test designed to evaluate the attention of children in Korea. The reliability and validity of CAT have been confirmed [20]. Simple selective attention (visual), simple selective attention (auditory), inhibition-sustained attention, and interference-selective attention proceed through CAT. The tool measures attention based on four factors: omission errors (inattention), commission errors (impulsivity), response time (processing speed), and response-time standard deviation (response-time variability).

3.4 Data Analysis

Compliance was assessed based on the percentage of each child performing the daily tasks. Therefore, the compliance trend was analyzed as the weekly rate of each child's daily task performance.

SPSS Statistics 25 was used for the data analysis. Reliability analysis was conducted on the self-efficacy and K-ARS subfactors. The reliability of the evaluation was considered appropriate when the Cronbach's alpha value was higher than 0.7 [11]. Normality tests were then performed. A repeated measures analysis of variance (rmANOVA) was also performed to identify the changes in ADHD symptoms and to explore self-efficacy awareness over three time points. If the sphericity assumption was not satisfied, the values for the Greenhouse-Geisser were inspected. Paired samples t-tests were used to compare CAT sub-factors before and after using the prototype. Statistical significance was defined as $p < 0.05$.

After transcription, the interview data were analyzed using an open-coding method based on grounded theory [24]. Two researchers constantly discussed and interpreted the data.

4 RESULTS

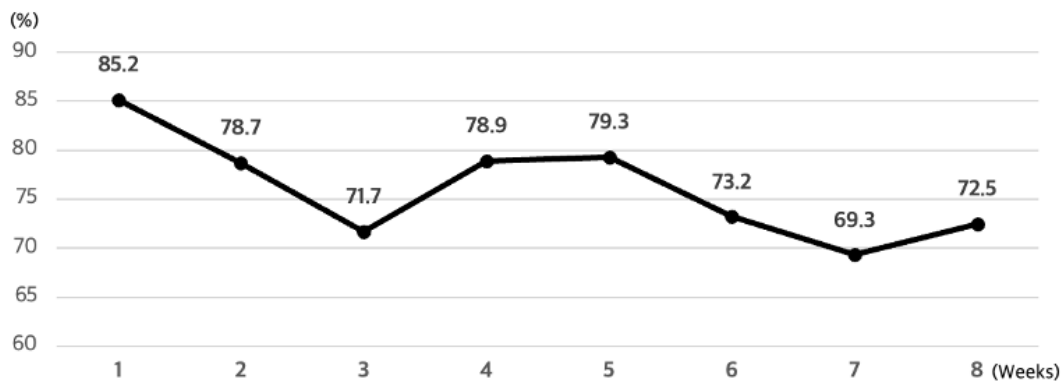
4.1 Participants

Ten children (four boys and six girls) who met the inclusion criteria participated in this study (Table 1). All participants had ADHD. Five

Table 1: Participants

Participant	Age	Gender	IQ	Division	Diagnosis ^a
P1	6	Girl	52	MID	ADHD
P2	6	Boy	62	MID	ADHD
P3	6	Boy	65	MID	ADHD, Dyslexia
P4	6	Girl	65	MID	ADHD
P5	6	Boy	69	MID	ADHD, Obsessive Compulsive Disorder
P6	7	Girl	70	BIF	ADHD
P7	7	Girl	80	BIF	ADHD
P8	7	Girl	81	BIF	ADHD
P9	6	Girl	81	BIF	ADHD
P10	6	Boy	84	BIF	ADHD

^a MID: Mild Intellectual Disability, BIF: Borderline Intellectual Functioning, ADHD: Attention deficit hyperactivity disorder

**Figure 2: Graph of average usage rate of prototypes of eight participants over eight weeks**

children had MID and five had BIF. All the children had an average IQ of 71.

Two out of the ten participants stopped using the prototype. P5 used the prototype for 29 days, and P4 used it for 17 days before requesting to discontinue its use. P5's mom said, "There are many holidays, like Christmas and New Year's Day, when my family has to go on vacation; therefore, it is hard to use the prototype" and "When my child had a 'Focus on studying' task, he needed a specific task like reading a book or writing in a diary. If the instructions aren't clear, he thinks it's something else." P4's mom also requested discontinuation, stating, "We live with the child's grandparents. As much as grandma helps with dressing my child, she has never done anything independently. However, all of a sudden, my child has to do something on her own; Thus, I guess that my child feels more comfortable not using the prototype." Therefore, in this study, an analysis was conducted on the data of eight children (three MID and five BIF).

4.2 Compliance

As shown in Figure 2, although the performance rate of daily tasks decreased over time, it showed a high performance rate of 76.1% on an average for eight weeks. P2 exhibited the highest performance rate. He achieved a daily task performance rate of 97.3% (214 tasks completed of 220 tasks to be done). In contrast, P3 showed the

lowest performance rate of 43.7% (73 tasks completed of 167 tasks to be done). Four children showed a 95% or higher performance rate for eight weeks (P1, P2, P6, and P7).

P1 explained that she used the prototype independently and performed the tasks. For instance, she said, "Even when my mom is sleeping, I can now get up on my own with the agent." Meanwhile, P2's mom said, "The kid who used to watch TV every day is now studying and organizing even when I don't speak." She was amazed by her child's task performance and changes in behavior. For P5, the usage rate for the prototype was low overall, but it improved from the seventh week. P5's mom said, "Until the middle, I thought my child was not good at programming; therefore, I thought we should quit. However, the child suddenly finds a tablet and performs daily tasks on his own. So, I thought my child was slow, but he could do it like other kids."

4.3 Self-efficacy

Cronbach's alpha values were as follows: Self-regulatory efficacy = 0.915, self-confidence = 0.724, and task difficulty preference = 0.827, indicating reliability. A normality test through Kolmogorov-Smirnov analysis was also performed.

Changes in the self-efficacy sub-factors are summarized in Table 2. The rmANOVA results indicated that the scores of 'task difficulty preference' ($F = 9.414, p = 0.003$), and 'total self-efficacy' ($F =$

Table 2: Changes in self-efficacy sub-factors over eight weeks

Sub-factors ^a	Visit 1 (0 weeks)	Visit 2 (4 weeks)	Visit 3 (8 weeks)	Visit 2-Visit 1	Visit 3-Visit 1	Visit 3-Visit 2	F / p
	M ± SD	M ± SD	M ± SD	Value / p	Value / p	Value / p	
SRE	2.95 ± 0.52	3.43 ± 0.40	3.78 ± 0.44	0.48 / 0.227	0.83 / 0.403	0.35 / 0.526	3.848 / 0.079
SCF	2.55 ± 0.20	3.28 ± 0.42	3.30 ± 0.53	0.73 / 0.409	0.75 / 0.345	0.03 / 1.000	2.316 / 0.135
TDP	1.93 ± 0.27	3.03 ± 0.43	3.83 ± 0.36	1.10 / 0.127	1.90 / 0.025	0.80 / 0.140	9.414 / 0.003
Total	2.48 ± 0.25	3.24 ± 0.28	3.63 ± 0.39	0.77 / 0.053	1.16 / 0.029	0.39 / 0.101	10.815 / 0.001

^a SRE: Self-regulatory Efficacy, SCF: Self-confidence, TDP: Task Difficulty Preference, Total: Total Self-efficacy

Table 3: Changes in K-ARS sub-factors over eight weeks

Sub-factors ^a	Visit 1 (0 weeks)	Visit 2 (4 weeks)	Visit 3 (8 weeks)	Visit 2-Visit 1	Visit 3-Visit 1	Visit 3-Visit 2	F / p
	M ± SD	M ± SD	M ± SD	Value / p	Value / p	Value / p	
I-A	9.50 ± 2.35	8.88 ± 1.79	7.00 ± 1.76	-0.63 / 1.000	-2.50 / 0.084	-1.88 / 0.105	4.487 / 0.031
H-I	8.13 ± 2.19	6.75 ± 1.69	5.13 ± 1.88	-1.38 / 0.518	-3.00 / 0.028	-1.63 / 0.498	5.129 / 0.021
Total	17.63 ± 4.44	15.63 ± 3.40	12.13 ± 3.58	-2.00 / 0.721	-5.50 / 0.016	-3.50 / 0.227	6.478 / 0.010

^a I-A: Inattention, H-I: Hyperactivity-Impulsivity, Total: Total ARS

10.815, $p = 0.001$) significantly increased over time. The results, to a marginally significant degree, demonstrated that self-regulatory efficacy ($F = 3.848$, $p = 0.079$) improved.

Regarding the preference for the task difficulty preference factor, it showed a statistically significant increase. As P1's mom said, "At first, my child only did things that were easy, like organizing her shoes, but now she is trying to do more difficult tasks, like studying," indicating that the child was taking on more tasks than before.

Although not statistically significant, the values of self-regulatory efficacy' and 'self-confidence generally increased. In an interview on week 8, P7 said, "I think I can do it alone! I think I can do a few tasks even without the agent." The children's confidence in self-expression seems to have been significantly enhanced by using the prototype. Furthermore, P1's mom said, "Previously, my child would stop trying to say something and then often say that it was nothing or that she forgot. Now, my child says everything that she wants to, maybe because she was talking to the agent frequently." Additionally, P2's mom said, "Previously, when I asked something, my child used to say he didn't know, but now he says at least a few words." We assume that the use of the prototype led to task execution and that this experience of achievement promoted self-efficacy.

4.4 K-ARS

The Cronbach's alpha values for this study were as follows: inattention = 0.923, hyperactivity-impulsivity = 0.895, indicating satisfactory reliability. A Kolmogorov-Smirnov analysis of the values of the factors confirmed that the sub-factors of the K-ARS satisfied the normality test.

All scores were significantly reduced at assessment compared to baseline (Table 3). The rmANOVA results indicated that the scores of inattention ($F = 4.487$, $p = 0.031$), hyperactivity-impulsivity ($F = 5.129$, $p = 0.021$), and total ARS ($F = 6.478$, $p = 0.010$) significantly reduced over time.

The mother of P6 said, "It was difficult to get my child to brush her teeth. But now she brushes her teeth and comes out on her own." P1's mom said, "My child didn't follow the rules at school. However, she tried to keep her promise to the agent at the scheduled time," indicating that P1 was trying to control her behavior. P2's mother said, "My child had a hard time waiting for someone to say or do something. Now he seems to know his turn to speak." That is, we guessed that children's inattention, hyperactivity, and impulsivity could be reduced using the prototype.

4.5 CAT

The CAT result was calculated and provided as an attention quotient (AQ), with a standardized average value of 100 and a standard deviation of 15. However, the AQ was not provided if it was less than 70. Because this was a study in which children with MID or BIF participated, many items were counted below 70.

As this study was designed as a pre-post experiment, the increase or decrease in each item was confirmed by a paired t-test for the raw CAT data. A normality test through Kolmogorov-Smirnov analysis was also performed. Table 4 summarizes the results of the commission errors, which were confirmed from four aspects. Commission errors are indicators of a child's impulsivity. With improvements in the data, improvements in children's impulsivity could be inferred. There was a significant difference in the 'inhibition-sustained attention' factor between the pretest ($M = 27.38$, $SD = 11.22$) and posttest ($M = 16.38$, $SD = 12.15$); $t(7) = 3.006$, $p = 0.020$. The statistical significance of the other factors was not confirmed.

5 DISCUSSION

The average compliance rate of children who completed using the prototype for eight weeks was confirmed as 76.1%. Half of the children had a compliance rate of 95% or higher. Although the children had a relatively low IQ, it was possible to assume that

Table 4: Changes in K-ARS sub-factors over eight weeks

Commission errors	Pretest M $\pm$ SD	Posttest M $\pm$ SD	<i>T</i>	<i>P</i>
Simple selective attention (Visual)	36.38 $\pm$ 23.51	28.88 $\pm$ 27.05	1.234	0.257
Simple selective attention (Auditory)	33.75 $\pm$ 21.84	19.75 $\pm$ 19.20	1.326	0.227
Inhibition-sustained attention	27.38 $\pm$ 11.22	16.38 $\pm$ 12.15	3.006	0.020
Interference-selective attention	33.00 $\pm$ 18.97	24.88 $\pm$ 19.80	0.796	0.452

they had no difficulty using the prototype. However, two children stopped using the prototype, which resulted in a dropout rate of 20%. As P4's mother said, it was not easy for a child with MID to try and do something independently. In addition, as the mother of P5 said, it was vital for children with MID to specify specific tasks to do. Therefore, future research should develop a method that uses more specific tasks that reflect the comments of P5's mother. Nevertheless, the high compliance shown by children over eight weeks can signify the conversational agent's potential to promote a positive outcome in children with MID.

Although not all sub-factors were statistically significant, self-efficacy increased overall. It was said that the P7 could do daily tasks alone without the prototype. The children's repeated use of the prototype and the performance and completion of daily tasks affected their self-efficacy. In addition, considering the cases of P1 and P2, it was confirmed that the conversational agent and prototype had a daily conversation, and self-expression about themselves increased. This can be interpreted as substantially positively affected children with MID.

Moreover, it was possible to estimate the improvement of children's detailed symptoms of ADHD using the prototype. Interpretation of the two indicators suggests that children's performance on the prototype and daily tasks influenced their impulsivity.

6 CONCLUSION

In this study, ten children with 1) ADHD and 2) MID or BIF were asked to perform daily tasks of their choice using a prototype at home for eight weeks. The prototype was equipped with a conversational agent that speaks based on self-instruction to help the child think and act. Eight out of ten children completed the eight-week experiment, with an average compliance rate of 76.1%. Children showed decreases related to K-ARS and increases related to self-efficacy. The interviews also revealed that their self-efficacy increased because of their increased self-expression. Furthermore, they desired to perform complex tasks and were willing to control their behavior. The interview data also showed improvements in ADHD symptoms and self-efficacy.

This study has a few limitations. First, as this was a small-scale study with only eight participants, the results cannot be generalized. Therefore, it is necessary to conduct further studies with more subjects to verify the current results. Second, two participants could not complete the study. Considering the characteristics of children with MID, they should be presented with specific tasks rather than broad task categories.

Nevertheless, this study is significant for several reasons. First, it was conducted for eight weeks, which enabled us to identify

changes in children's behavior. We observed the experiences of low cognitive ability children with ADHD using prototypes. Second, each child's self-efficacy and ADHD symptoms were analyzed based on a survey, interview, and computerized tool to investigate the implications of the outcomes. Although this was a pilot study, it was possible to estimate the effectiveness of the prototype by checking various types of data.

The results of this study indicate that conversational agents can improve the quality of life of children with neurodevelopmental disorders such as ADHD and ID. In future research, we plan to explore whether conversational agents can positively affect the quality of life of typically developing children through analysis of digital phenotyping data such as voice and HRV.

ACKNOWLEDGMENTS

This research was supported by a grant of the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health & Welfare, Republic of Korea (grant number: HI22C0775). And this work was partly supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (NRF- 2016R1D1A1B02015987).

REFERENCES

- [1] Aisha Holloway RGN and Hazel E Watson RGN. 2002. Role of self-efficacy and behaviour change. *International journal of nursing practice* 8, 2, 106-115. <https://doi.org/10.1046/j.1440-172x.2002.00352.x>
- [2] Albert Bandura. 1992. Exercise of personal agency through the self-efficacy mechanism. *Self-efficacy: Thought control of action* 1, 3-37.
- [3] Albert Bandura. 1995. Exercise of personal and collective efficacy in changing societies. *Self-efficacy in changing societies* 15, 334.
- [4] Albert Bandura. 1997. *Self-efficacy: The Exercise of Control*. New York, NY: Freeman.
- [5] American Psychiatric Association. 2013. *Diagnostic and statistical manual of mental disorders: DSM-5*. American Psychiatric Pub. Washington, DC
- [6] Anna Jankowska. 2016. Towards a framework for psychological resilience in children and adolescents with Borderline Intellectual Functioning. *Polish Psychological Bulletin*. 47, 3, 289–299. <https://doi.org/10.1515/ppb-2016-0035>
- [7] Annalisa Valle, Gisella Baglio, Michela Zanette, Davide Massaro, Francesca Baglio, Antonella Marchetti, and Valeria Blasi. 2021. A New Perspective on the Role of Self-Confidence and Confidence in the Evaluation and Rehabilitation of Children With Adverse Life Experience and Borderline Intellectual Functioning: A Preliminary Study. *Frontiers in Psychology*, 12. <https://doi.org/10.3389/fpsyg.2021.720219>
- [8] Ariana M. Albanese, Gabrielle R. Russo, and Pamela A. Geller. 2019. The role of parental self-efficacy in parent and child well-being: A systematic review of associated outcomes. *Child: care, health and development* 45, 3, 333-363. <https://doi.org/10.1111/cch.12661>
- [9] Barry J. Zimmerman. 1995. Self-efficacy and educational development. *Self-efficacy in changing societies* 1, 1, 202-231.
- [10] Cameron L. Neece, Bruce L. Baker, Keith Crnic, and Jan Blacher. 2013. Examining the validity of ADHD as a diagnosis for adolescents with intellectual disabilities: Clinical presentation. *Journal of Abnormal Child Psychology* 41, 4, 597-612. <https://doi.org/10.1007/s10802-012-9698-4>
- [11] Cortina, Jose M. 1993. What is coefficient alpha? An examination of theory and applications. *Journal of applied psychology*. 78, 1, 98–104. <https://doi.org/10.1037/>

- 0021-9010.78.1.98
- [12] Do Eun Park, Yee Jin Shin, EunAh Park, In Ae Choi, Woo Yeon Song, and Jinwoo Kim. 2020. Designing a Voice-Bot to Promote Better Mental Health; UX Design for Digital Therapeutics on ADHD Patients. Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI'20). Association for Computing Machinery, Honolulu, HI, USA, 1-8. <https://doi.org/10.1145/3334480.3382948>
 - [13] Do Eun Park, Junghan Lee, Jeehyun Han, Jinwoo Kim, and Yee Jin Shin. 2023. A Preliminary Study of Voicebot to Assist ADHD Children in Performing Daily Tasks. *International Journal of Human-Computer Interaction*, 1-14. <https://doi.org/10.1080/10447318.2023.2169530>
 - [14] Dobson, Keith S. and Dozois, David J. A. 2010. Historical and philosophical bases of the cognitive-behavioral therapies. In K. S. Dobson (Ed.), *Handbook of cognitive-behavioral therapies* (pp. 3–38). Guilford Press.
 - [15] Doeun Park, Myounglee Choo, Jinwoo Kim, Junghan Lee, and Yee-Jin Shin. 2022. Conversational Agent for Creating Regularity in Children with ADHD: A Quantitative and Qualitative Pilot Study. In *Adjunct Publication of the 24th International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI'22)*. Association for Computing Machinery, Vancouver, BC, Canada, 1-6. <https://doi.org/10.1145/3528575.3551452>
 - [16] Donald H. Meichenbaum and Joseph Goodman. 1971. Training impulsive children to talk to themselves: a means of developing self-control. *Journal of abnormal psychology* 77, 2(April 1971), 115-126.
 - [17] DuPaul, George J. Power, Thomas J. Anastopoulos, and Arthur D. Reid, Robert. 1998. *ADHD Rating Scale—IV: Checklists, norms, and clinical interpretation*. Guilford press.
 - [18] Éva Bócsa. 2003. School adjustment of borderline intelligence pupils. *Strategies* 44, 8.
 - [19] Gloria E. Miller. 1987. The influence of self-instruction on the comprehension monitoring performance of average and above average readers. *Journal of Reading Behavior* 19, 3. 303-317. <https://doi.org/10.1080/10862968709547606>
 - [20] Hanik K. Yoo, Jungsun Lee, Sung Hee Kang, Eun Hee Park, Jaesuk Jung, Boong-Nyun Kim, Jung-Woo Son, Tae Won Park, Bongseok Kim, and Young Sik Lee. 2009. Standardization of the Comprehensive Attention Test for the Korean Children and Adolescents. *Journal of the Korean Academy of Child and Adolescent Psychiatry* 20, 2. 68-75.
 - [21] Henriette Michalsen, Silje Camilla Wangberg, Gunnar Hartvigsen, André Henriksen, Gunn Pettersen, Letizia Jaccheri, Reidun Jahnsen, Gyrð Thrane, Cathrine Arntzen, and Audny Anke. 2022. A. Mobile health support to stimulate physical activity in individuals with intellectual disability: Protocol for mixed methods pilot study. *JMIR Research Protocols* 11, 9. e37849. <https://doi.org/10.2196/37849>
 - [22] Henrik Danielsson, Lucy Henry, Jerker Rönnberg, and Lars-Göran Nilsson. 2010. Executive functions in individuals with intellectual disability. *Research in Developmental Disabilities* 31, 6. 1299-1304. <https://doi.org/10.1016/j.ridd.2010.07.012>
 - [23] James B. Wise. 2002. Social cognitive theory: A framework for therapeutic recreation practice. *Therapeutic Recreation Journal* 36, 4. 335-351.
 - [24] Juliet Corbin and Anselm Strauss. 1990. Grounded theory research: Procedures, canons, and evaluative criteria. *Qualitative sociology* 13, 1 (March 1990). 3-21. 2010.
 - [25] Katherine A. Smith, Kevin A. Ayres, Jennifer Alexander, Jennifer R. Ledford, Collin Shepley and Sally B. Shepley. 2016. Initiation and generalization of self-instructional skills in adolescents with autism and intellectual disability. *Journal of Autism and Developmental Disorders* 46. 1196–1209. <https://doi.org/10.1007/s10803-015-2654-8>
 - [26] Lykke Brogaard Bertel, Helene Lassen Nørlem, and Melissa Azari. 2020. Supporting Self-Efficacy in Children with ADHD through AI-supported Self-monitoring: Initial Findings from a Case Study on Tiimood. In *Adjunct proceedings of the 15th International Conference on Persuasive Technology*. 1-4
 - [27] Mark Sherer, James E. Maddux, Blaise Mercandante, Steven Prentice-Dunn, Beth Jacobs, and Ronald W. Rogers. 1982. The Self-Efficacy Scale: Construction and Validation. *Psychological reports* 51, 2. 663-671. <https://doi.org/10.2466/pr0.1982.51.2.663>
 - [28] Michael Ferrari. 2009. Borderline intellectual functioning and the intellectual disability construct. *Intellectual and developmental disabilities*, 47, 5. 386–389. <https://doi.org/10.1352/1934-9556-47.5.386>
 - [29] Petri J.C.Membregts. 2000. Effectiveness of video feedback and self-management on inappropriate social behavior of youth with mild mental retardation. *Research in Developmental Disabilities* 21, 5. 409-423. [https://doi.org/10.1016/S0891-4222\(00\)00052-4](https://doi.org/10.1016/S0891-4222(00)00052-4)
 - [30] Ratna Narayan, and David Lamp. 2010. "Me? Teach science?" Exploring EC-4 pre-service teachers' self efficacy in an inquiry-based constructivist physics classroom. *Educational Research and Reviews* 5, 12. 748-757.
 - [31] Rhoda Cummings, Cleborne D. Maddux, and Jack Casey. 2000. Individualized transition planning for students with learning disabilities. *The Career Development Quarterly* 49, 1. 60-72. <https://doi.org/10.1002/j.2161-0045.2000.tb00751.x>
 - [32] Seong-sook U, and Jin-Sook Shin. 2013. A Effect of UCC Making Activity on Self-efficacy and Self-directed Learning Ability of Children with Intellectual Disabilities. *The Journal of Special Education: Theory and Practice* 14, 1. 235-257.
 - [33] Sophie Leitch, Emma Sciberras, Brittany Post, Bibi Gerner, Nicole Rinehart, Jan M. Nicholson and Subhadra Evans. 2019. Experience of stress in parents of children with ADHD: A qualitative study. *International journal of qualitative studies on health and well-being* 14, 1. 1690091. <https://doi.org/10.1080/17482631.2019.1690091>
 - [34] Stephanie L Haft, Chelsea A Myers, and FumikoHoeft. 2016. Socio-emotional and cognitive resilience in children with reading disabilities. *Current opinion in behavioral sciences* 10, 1. 133-141. <https://doi.org/10.1016/j.cobeha.2016.06.005>
 - [35] Sunil Karande, Sandeep Kanchan, and Madhuri Kulkarni. 2008. Clinical and psychoeducational profile of children with borderline intellectual functioning. *The Indian Journal of Pediatrics* 75, 8. 795-800. <https://doi.org/10.1007/s12098-008-0101-y>
 - [36] Susan Young, and J. Myanathi Amarasinghe. 2010. Practitioner Review: Non-pharmacological treatments for ADHD: A lifespan approach. *Journal of Child Psychology and Psychiatry* 51, 2. 116-133. <https://doi.org/10.1111/j.1469-7610.2009.02191.x>
 - [37] Thomas E. Brown. 2009. ADD/ADHD and impaired executive function in clinical practice. *Current Attention Disorders Reports* 1, 1. 37-41. <https://doi.org/10.1007/s12618-009-0006-3>
 - [38] Torgesen, Joseph K. 1994. Issues in the assessment of executive function: An information-processing perspective. In G. R. Lyon (Ed.), *Frames of reference for the assessment of learning disabilities: New views on measurement issues* (pp. 143–162). Paul H Brookes Publishing Co..
 - [39] Yu-Kyoung So, Joo-Sun Noh, Young-Shin Kim, Seon-Gyu Ko, and Yun-Joo Koh. 2002. The reliability and validity of Korean parent and teacher ADHD rating scale. *Journal of Korean Neuropsychiatric Association* 41, 2. 283-289.